

Optical Flow Modulation: Counteracting Physical and Apparent Motion to Regulate Motion Impression

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ABSTRACT

We propose a visual cueing method termed luminance mixing, inspired by apparent motion, which induces motion impression through temporal luminance modulation across images captured from multiple static viewpoints, without geometric displacement. We examined whether luminance mixing produces optical-flow-like signals and whether such signals, when opposing those from physical motion, regulate perceived motion impression. Optical-flow was evaluated via participants' subjective self-motion impression reports. Results showed that luminance mixing reliably elicited motion impression between static images and attenuated motion impression induced by physical motion, demonstrating control over motion impression.

Index Terms: Optical Flow, Motion Impression, Physical Motion, Apparent Motion, Temporal Luminance Modulation, Cancellation

1 INTRODUCTION

Visual motion impression plays a critical role in human activities such as vehicle operation and is primarily driven by optical flow—the structured retinal motion produced by relative movement between an observer and the environment. Beyond motion arising from physical displacement, the visual system can also perceive motion through apparent motion, in which sequential presentation of static images induces a motion impression. Previous studies have shown that apparent motion, including motion induced solely by temporally modulated luminance, can generate optical-flow-like signals sufficient to support higher-level spatial perception [1] [2].

Building on these findings, the present study investigates whether motion induced by temporally modulated luminance under static viewpoints can produce motion impressions comparable to physical motion, and whether luminance-induced motion oriented opposite to physical optical flow can attenuate perceived motion without altering scene geometry. Since excessive or conflicting motion impression is a known contributor to motion sickness, the ability to suppress perceived motion through luminance mixing suggests potential applications for discomfort reduction in vehicular and VR environments.

2 METHOD

2.1 Luminance Mixing

The luminance mixing method combines images captured from multiple discrete, fixed cameras and modulates their luminance contributions over time. All cameras remain spatially static; no physical viewpoint movement is introduced. Motion-related visual cues are instead generated by temporally varying the relative luminance of images from different viewpoints. A simple temporal switching between camera images produces only a discontinuous alternation of static views and does not provide continuous visual

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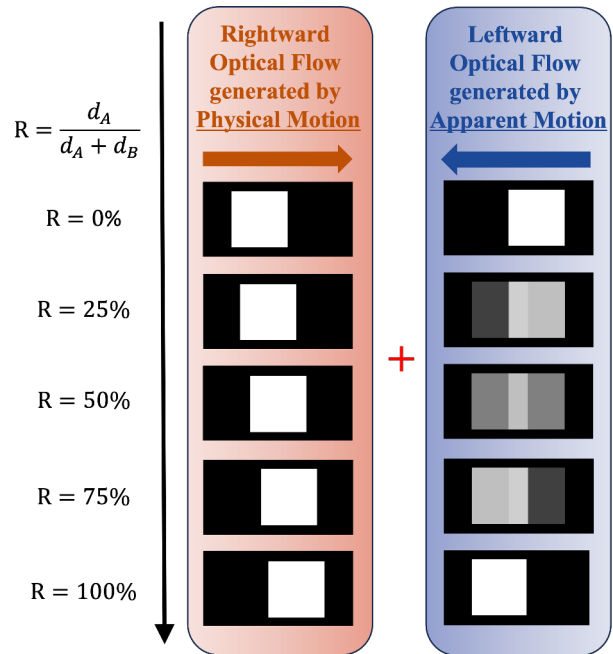


Figure 1: Superposition of Physical Motion and Apparent Motion. The parameter R represents the relative position of Camera C between Camera A and Camera B and is computed using inverse distance weighting. This parameter determines the luminance contribution of Image B during apparent motion generation. Videos corresponding to physical motion and apparent motion are then temporally combined and presented to the participant.

cues between observation points. In contrast, smoothly varying the luminance contributions of multiple images over time yields a continuous change in the overall luminance distribution.

Each camera image contains a distinct luminance pattern. When these patterns are modulated continuously over time, optical-flow-like signals can arise on the retina despite the absence of physical spatial motion. For example, a luminance-mixed image can be formulated as a distance-weighted combination of images from different viewpoints. Through this time-based luminance mixing, continuous luminance transitions are achieved without introducing geometric displacement.

2.2 Procedure

The experiment employed a within-subjects design in which all participants experienced the same four experiments. Six participants took part in the study, and each participant completed six trials per condition. The first trial served as a practice trial and was excluded from analysis, while the remaining five trials were averaged to obtain a representative luminance mixing amplitude for each participant and condition. At the beginning of each trial, participants pressed the Enter key to initiate stimulus presentation. During the

trial, participants were instructed to adjust the luminance mixing modulation amplitude, rather than the physical motion amplitude, to either maximize or minimize the perceived motion impression. The modulation amplitude was defined as the absolute value within the range of $[0, 1]$, where 1 corresponded to the maximum perceived motion and 0 to the minimum. To avoid edge bias, the adjustment range in Experiment 4 was extended to $[-1, 1]$, with the Cancellation condition centered at 0. The modulation amplitude could be freely and repeatedly adjusted using a rotary encoder. No time limit was imposed, and each trial ended once the participant reached a stable perceptual judgment.

Table 1: Experimental Conditions

Exp	Condition
1	Static Image vs. Luminance Mixing
2	Physical Motion vs. Cancellation
3	Double Image vs. Cancellation
4	Physical Motion vs. Cancellation vs. Double Image

Five visual stimulus conditions were used in the experiment. The Static Image condition presented a single static image captured by a fixed camera. In the Luminance Mixing condition, apparent motion was generated through temporal modulation of the luminance ratio between images captured by two fixed cameras separated by 0.25 m, with luminance mixing performed at a frequency of 50 Hz. The Physical Motion condition involved visual input captured by a camera undergoing periodic left–right oscillatory motion at 50 Hz. In the Double Image condition, images from two cameras separated by 0.25 m were combined with equal luminance contributions (50% each) and presented with periodic oscillation at 50 Hz. Finally, the Cancellation condition combined physical motion with apparent motion in the opposite direction (Fig. 1).

2.3 System

The visual stimulus generation system was implemented in Unity 2022.3.30f1 (LTS). All stimuli were rendered at a frame rate of 240 fps with a spatial resolution of 2560×1440 pixels to ensure smooth temporal luminance modulation and to minimize temporal aliasing. The virtual cameras in the environment had a field of view of 60° , and participants maintained a forward-facing posture toward the center of the display at the same height, with a fixed viewing distance of 0.5 m.

To enable real-time control of luminance modulation amplitude, an M5Stack rotary encoder unit was integrated as an input device. The rotary encoder provided continuous analog input, which was transmitted to the Unity application in real time and mapped directly to the luminance mixing parameters, allowing interactive adjustment of stimulus intensity during presentation.

3 RESULTS

Across all experimental conditions, the findings were consistent with the expected effects of luminance mixing on perceived motion impression (Fig. 2). Luminance mixing between static images reliably induced a strong motion impression compared to a single static image, whereas cancellation-based luminance mixing consistently produced the weakest motion impression, effectively suppressing motion cues induced by physical camera motion. Notably, even in Experiment 4, where the adjustment range was extended to avoid edge bias, the Cancellation condition remained close to zero, indicating robust suppression of perceived motion. In contrast, the Physical Motion and Double Image conditions yielded the strongest perceived motion. Quantitatively, across all four experiments, modulation amplitudes in the Max condition were significantly larger than those in the Min condition (all $p < 10^{-5}$; $\alpha = 0.05$), with

extremely large effect sizes ($d_z = 7.07\text{--}33.23$). Mean amplitudes were close to 1.0 for Max conditions (0.958–1.000) and near zero for Min conditions (0.026–0.195). Together, these results demonstrate that perceived motion impression can be systematically and robustly enhanced or suppressed through luminance mixing parameters, with optical-flow-like signals enhancing motion impression and opposing luminance-driven motion reliably attenuating it.

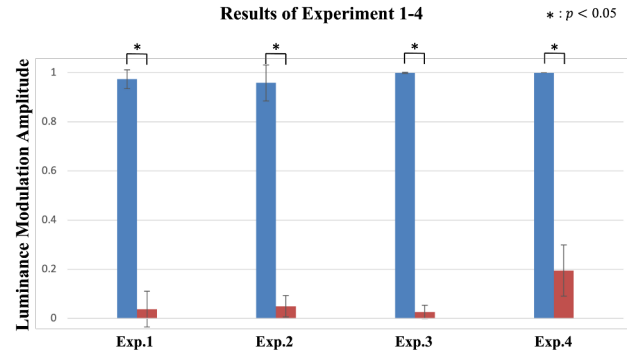


Figure 2: Results of Experiment 1-4. Results of Experiments 1–4. The modulation amplitude was defined as the absolute value in the range of 0–1, where 1 corresponds to the maximum perceived motion and 0 to the minimum. Instructions: (Blue Bar) Maximize the Perceived Motion Impression; (Red Bar) Minimize the Perceived Motion Impression.

4 CONCLUSION

This study proposed a method termed luminance mixing and demonstrated that temporal luminance modulation can effectively regulate perceived motion impression. Moreover, when motion induced by luminance mixing is applied in the direction opposite to physical motion, it can suppress motion impression induced by physical motion. Across all experimental conditions except the Static Image condition, the Cancellation condition consistently produced the weakest perceived motion impression, whereas the Physical Motion and Double Image conditions yielded the strongest motion impression. These findings highlight luminance mixing as a flexible mechanism for shaping motion impression through temporal luminance modulation. By decoupling perceived motion from physical camera movement, the proposed approach enables controlled induction or suppression of motion impression at the perceptual level and provides a foundation for future studies on perceptually driven motion modulation. Because motion sickness has been linked to mismatches and exaggerations in perceived self-motion, this capability suggests potential applications for discomfort reduction in vehicular and VR environments. Future work will directly examine whether cancellation-based luminance mixing leads to measurable reductions in motion sickness symptoms.

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